

PRAKTIKUM BASIS DATA

MODUL 8

(08_Scripting_SQL_Create_)



Oleh:

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JURNAL

1. Soal 1

Jawaban:

```
CREATE TABLE department (  
    department_id NUMBER(5) PRIMARY KEY,  
    department_name VARCHAR2(50),  
    manager_id NUMBER(5),  
    location_id NUMBER(5)  
);  
  
CREATE TABLE supplier (  
    supplier_id NUMBER(5) PRIMARY KEY,  
    supplier_name VARCHAR2(100),  
    phone VARCHAR2(20),  
    address VARCHAR2(100)  
);  
  
CREATE TABLE employee (  
    employee_id NUMBER(5) PRIMARY KEY,  
    last_name VARCHAR2(50) NOT NULL,  
    email VARCHAR2(100),  
    salary NUMBER(10,2),  
    commission_pct NUMBER(5,2),  
    hire_date DATE NOT NULL,  
    department_id NUMBER(5),  
  
    CONSTRAINT employee_fk  
    FOREIGN KEY (department_id)  
    REFERENCES department(department_id)  
);
```

```

CREATE TABLE inventory (
    inventory_id NUMBER(5) PRIMARY KEY,
    item_name VARCHAR2(100),
    stock NUMBER(5),
    price NUMBER(10,2),
    supplier_id NUMBER(5),

    CONSTRAINT inventory_fk
    FOREIGN KEY (supplier_id)
    REFERENCES supplier(supplier_id)
);

```

2. Soal 2

Jawaban:

```

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;

```

Melakukan analisis statistik pada seluruh tabel menggunakan perintah ANALYZE TABLE COMPUTE STATISTICS agar Oracle dapat mengetahui jumlah row dan block pada tabel untuk optimasi query.

3. Soal 3

Jawaban:

```

INSERT INTO department VALUES (10, 'HR', 101, 1001);
INSERT INTO department VALUES (20, 'Finance', 102, 1002);
INSERT INTO department VALUES (30, 'IT', 103, 1003);
INSERT INTO department VALUES (40, 'Marketing', 104, 1004);
INSERT INTO department VALUES (50, 'Sales', 105, 1005);
INSERT INTO department VALUES (60, 'Production', 106, 1006);
INSERT INTO department VALUES (70, 'Logistic', 107, 1007);
INSERT INTO department VALUES (80, 'Operations', 108, 1008);
INSERT INTO department VALUES (90, 'Security', 109, 1009);
INSERT INTO department VALUES (100, 'Support', 110, 1010);

```

```

INSERT INTO supplier VALUES (1, 'PT Sumber Jaya', '0811111111', 'Bandung');
INSERT INTO supplier VALUES (2, 'PT Makmur', '0822222222', 'Jakarta');
INSERT INTO supplier VALUES (3, 'PT Abadi', '0833333333', 'Surabaya');
INSERT INTO supplier VALUES (4, 'PT Sentosa', '0844444444', 'Semarang');
INSERT INTO supplier VALUES (5, 'PT Maju', '0855555555', 'Bekasi');
INSERT INTO supplier VALUES (6, 'PT Karya', '0866666666', 'Bogor');
INSERT INTO supplier VALUES (7, 'PT Sejahtera', '0877777777', 'Depok');
INSERT INTO supplier VALUES (8, 'PT Nusantara', '0888888888', 'Yogyakarta');
INSERT INTO supplier VALUES (9, 'PT Cemerlang', '0899999999', 'Medan');
INSERT INTO supplier VALUES (10, 'PT Prima', '0800000000', 'Bali');

INSERT INTO employee VALUES (1, 'Andi', 'andi@gmail.com', 5000000, 0.1, TO_DATE('2023-01-01','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (2, 'Budi', 'budi@gmail.com', 6000000, 0.2, TO_DATE('2023-01-02','YYYY-MM-DD'), 20);
INSERT INTO employee VALUES (3, 'Citra', 'citra@gmail.com', 5500000, 0.1, TO_DATE('2023-01-03','YYYY-MM-DD'), 30);
INSERT INTO employee VALUES (4, 'Doni', 'doni@gmail.com', 7000000, 0.3, TO_DATE('2023-01-04','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (5, 'Eka', 'eka@gmail.com', 4500000, 0.1, TO_DATE('2023-01-05','YYYY-MM-DD'), 50);
INSERT INTO employee VALUES (6, 'Fajar', 'fajar@gmail.com', 5200000, 0.2, TO_DATE('2023-01-06','YYYY-MM-DD'), 60);
INSERT INTO employee VALUES (7, 'Gina', 'gina@gmail.com', 4800000, 0.1, TO_DATE('2023-01-07','YYYY-MM-DD'), 70);
INSERT INTO employee VALUES (8, 'Hadi', 'hadi@gmail.com', 5100000, 0.2, TO_DATE('2023-01-08','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (9, 'Intan', 'intan@gmail.com', 6500000, 0.3, TO_DATE('2023-01-09','YYYY-MM-DD'), 90);
INSERT INTO employee VALUES (10, 'Joko', 'joko@gmail.com', 4700000, 0.1, TO_DATE('2023-01-10','YYYY-MM-DD'), 100);

```

```

INSERT INTO inventory VALUES (1, 'Keyboard', 20, 250000, 1);

INSERT INTO inventory VALUES (2, 'Mouse', 30, 150000, 2);

INSERT INTO inventory VALUES (3, 'Monitor', 15, 2000000, 3);

INSERT INTO inventory VALUES (4, 'CPU', 10, 5000000, 4);

INSERT INTO inventory VALUES (5, 'Printer', 8, 1500000, 5);

INSERT INTO inventory VALUES (6, 'Scanner', 5, 1200000, 6);

INSERT INTO inventory VALUES (7, 'Speaker', 25, 300000, 7);

INSERT INTO inventory VALUES (8, 'Laptop', 12, 7000000, 8);

INSERT INTO inventory VALUES (9, 'Projector', 6, 3500000, 9);

INSERT INTO inventory VALUES (10, 'Router', 18, 800000, 10);

```

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	HR	101	1001
2	20	Finance	102	1002
3	30	IT	103	1003
4	40	Marketing	104	1004
5	50	Sales	105	1005
6	60	Production	106	1006
7	70	Logistic	107	1007
8	80	Operations	108	1008
9	90	Security	109	1009
10	100	Support	110	1010

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	1 Andi	andi@gmail.com	5000000	0.1	01-JAN-23	80
2	2 Budi	budi@gmail.com	6000000	0.2	02-JAN-23	20
3	3 Citra	citra@gmail.com	5500000	0.1	03-JAN-23	30
4	4 Doni	doni@gmail.com	7000000	0.3	04-JAN-23	80
5	5 Eka	eka@gmail.com	4500000	0.1	05-JAN-23	50
6	6 Fajar	fajar@gmail.com	5200000	0.2	06-JAN-23	60
7	7 Gina	gina@gmail.com	4800000	0.1	07-JAN-23	70
8	8 Hadi	hadi@gmail.com	5100000	0.2	08-JAN-23	80
9	9 Intan	intan@gmail.com	6500000	0.3	09-JAN-23	90
10	10 Joko	joko@gmail.com	4700000	0.1	10-JAN-23	100

SUPPLIER_ID	SUPPLIER_NAME	PHONE	ADDRESS
1	1 PT Sumber Jaya	0811111111	Bandung
2	2 PT Makmur	0822222222	Jakarta
3	3 PT Abadi	0833333333	Surabaya
4	4 PT Sentosa	0844444444	Semarang
5	5 PT Maju	0855555555	Bekasi
6	6 PT Karya	0866666666	Bogor
7	7 PT Sejahtera	0877777777	Depok
8	8 PT Nusantara	0888888888	Yogyakarta
9	9 PT Cemerlang	0899999999	Medan
10	10 PT Prima	0800000000	Bali

INVENTORY_ID	ITEM_NAME	STOCK	PRICE	SUPPLIER_ID
1	1 Keyboard	20	250000	1
2	2 Mouse	30	150000	2
3	3 Monitor	15	2000000	3
4	4 CPU	10	5000000	4
5	5 Printer	8	1500000	5
6	6 Scanner	5	1200000	6
7	7 Speaker	25	300000	7
8	8 Laptop	12	7000000	8
9	9 Projector	6	3500000	9
10	10 Router	18	800000	10

4. Soal 4

Jawaban:

```
ANALYZE TABLE department COMPUTE STATISTICS;  
ANALYZE TABLE employee COMPUTE STATISTICS;  
ANALYZE TABLE supplier COMPUTE STATISTICS;  
ANALYZE TABLE inventory COMPUTE STATISTICS;
```

Setelah data dimasukkan ke dalam tabel, dilakukan kembali analisis tabel menggunakan COMPUTE STATISTICS agar Oracle memperbarui statistik jumlah data pada setiap tabel.

5. Soal 5

Jawaban:

```
CREATE VIEW empvu80 AS  
SELECT employee_id AS id_number,  
       last_name AS name,  
       salary,  
       department_id  
FROM employee  
WHERE department_id = 80;
```

Membuat view empvu80 untuk menampilkan data pegawai yang bekerja pada department 80

6. Soal 6

Jawaban:

```
DESC empvu80;
```

Menampilkan struktur view empvu80 menggunakan perintah DESC

7. Soal 7

Jawaban:

```
CREATE INDEX idx_employee_department  
ON employee(department_id);
```

Membuat non-unique index pada kolom department_id di tabel employee untuk meningkatkan performa pencarian dan join pada foreign key.